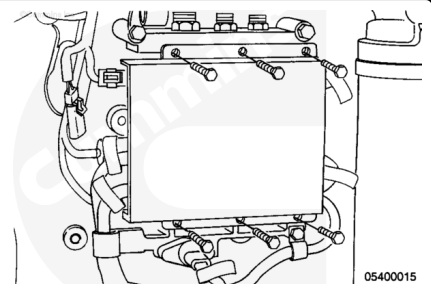


Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

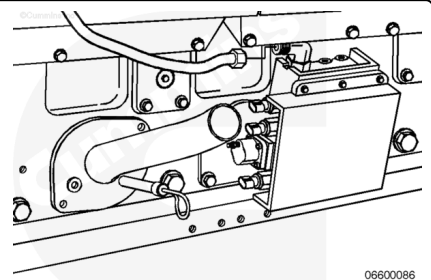
Remove the electronic control module (ECM). Refer to Procedure 019-031 (</qs3/pubsys2/xml/en/procedures/19/19-019-031.html>) in the Troubleshooting and Repair Manual, Electronic Control Systems, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.



Remove

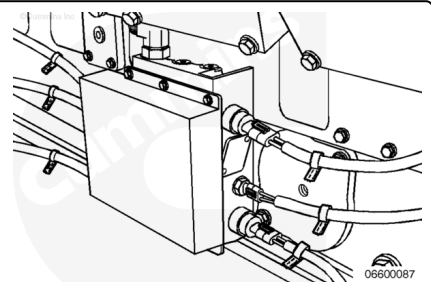
with Mechanically Actuated Injector

Clean the control valve body and the surrounding area.
Disconnect the fuel supply line at the control valve body.

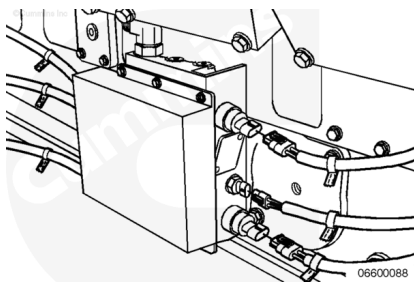


It is a good service practice to label parts during removal.
Identification of the connections at the time of removal with a label will aid in the installation process.

Label the electrical connections to the actuators, fuel pressure sensors, fuel temperature sensor, and ambient air pressure sensor.

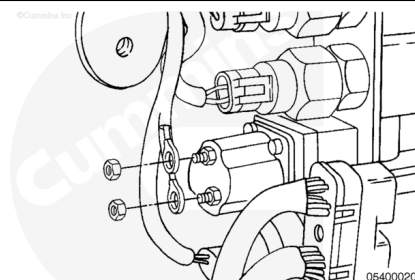


Disconnect the actuators and pressure sensor connections.



Use a 3/8-inch wrench to remove the nut holding the electrical connection of the fuel shutoff valve coil.

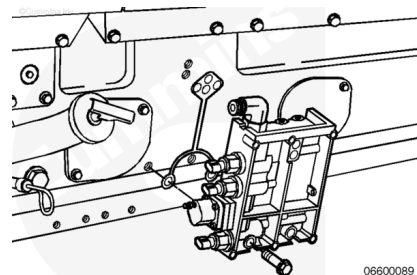
Remove the connector.



Move the wiring harness away from the control valve body to gain access, as needed.

Remove the control valve body assembly mounting capscrews, control valve body assembly, and gasket.

The fuel control valve body assembly gasket is an edge molded gasket. Inspect the gasket. If any damage is found it **must** be replaced.



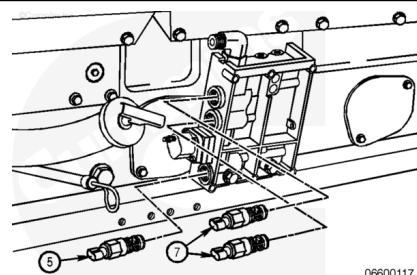
Disassemble

with Mechanically Actuated Injector

Use a 1¼-inch deep well socket to remove and install both actuators.

Remove the two fuel timing rail actuators (7).

Remove the fueling rail actuator (5).

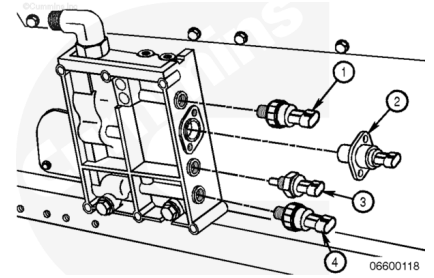


Use a 1¼-inch deep well socket to remove the following sensors.

- Fuel timing rail pressure sensor (1)
- Fuel temperature sensor (3)

- Fuel rail pressure sensor (4).

Do **not** use a 1¼ inch deep well socket to remove the Ambient Air Pressure Sensor (2) (if installed on the electronic fuel control valve assembly). Damage to the sensor will result.

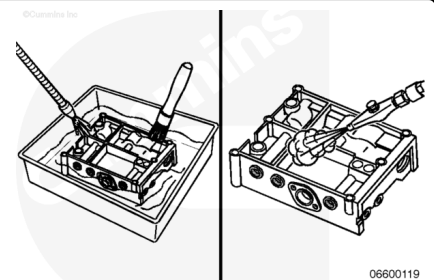


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



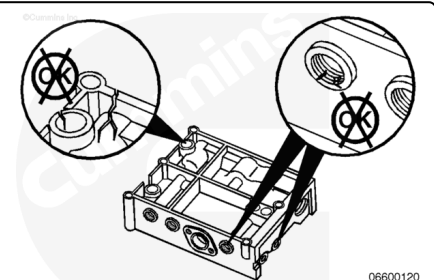
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the fuel control valve body.

Dry with compressed air.

Inspect the fuel control valve body sensor mounting areas for damaged threads and cracks in the housing.

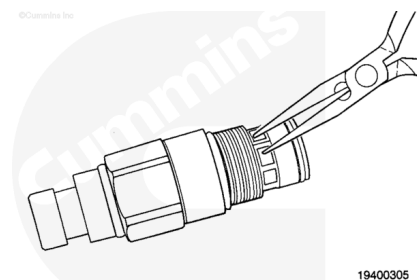


Inspect the actuator threads for damage.

Inspect the actuator screens for debris and damage.

Make sure the screen filter is in its groove and the two ends of the screen are nearly joined.

Using a pair of needle nose pliers, carefully join the screen filter ends.



Assemble

with Mechanically Actuated Injector

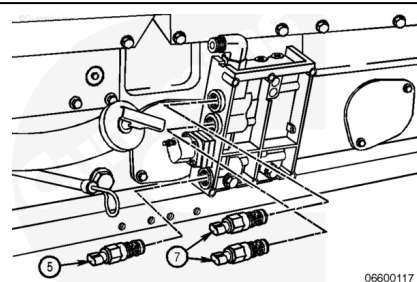
Use a 1¼-inch deep well socket to install both actuators.

Install the two timing actuators (7).

Torque Value: 25 n•m [18 ft-lb]

Install the fueling rail actuator (5).

Torque Value: 25 n•m [18 ft-lb]



Use a 1¼-inch deep well socket to install the following sensors except the ambient air pressure sensor.

Install the timing rail pressure sensor (1).

Torque Value: 14 n•m [124 in-lb]

Install the fuel temperature sensor (3).

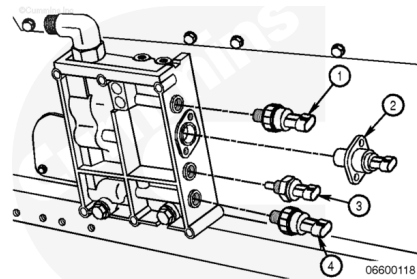
Torque Value: 15 n•m [133 in-lb]

Install the fuel rail pressure sensor (4).

Torque Value: 14 n•m [124 in-lb]

Do **not** use a 1¼ inch deep well socket to Install the Ambient Air Pressure Sensor (2) (if installed on the electronic fuel control valve assembly). Damage to the sensor will result.

Torque Value: 23 n•m [17 ft-lb]



Install

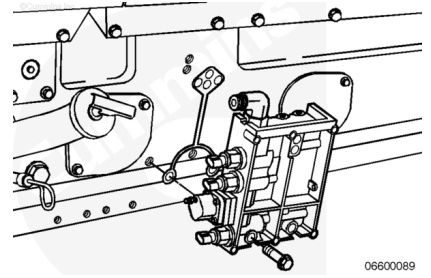
with Mechanically Actuated Injector

Install the fuel control valve body gasket.

Install the fuel control valve body assembly and mounting capscrews.

Tighten the mounting capscrews.

Torque Value: 45 n•m [33 ft-lb]

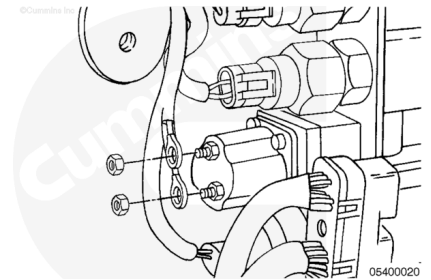


Install the electrical connection on the fuel shutoff valve.

Install the nut on the threaded post of the coil.

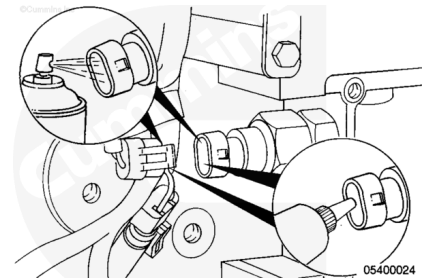
Use two 3/8-inch wrenches to hold the post of the nut firmly while tightening the connection nut.

Torque Value: 2 n•m [18 in-lb]



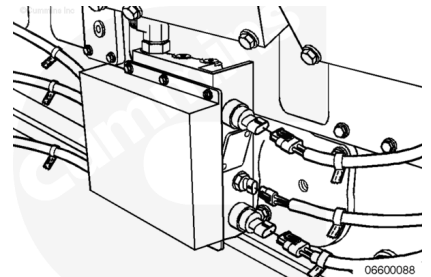
Inspect the actuator and pressure sensor connections for dirt and debris.

Use QD contact cleaner, Part Number 3824510, or equivalent, to clean the connections and actuator, as needed.

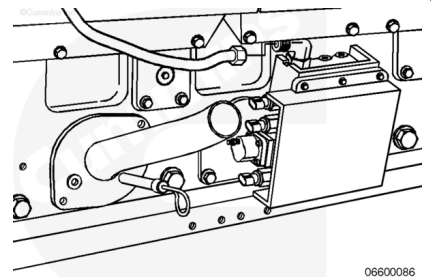


If the electrical connections were **not** labeled for identification purposes during disassembly, verify the types of connections on the wiring diagram.

Connect the actuator and pressure sensor connections.



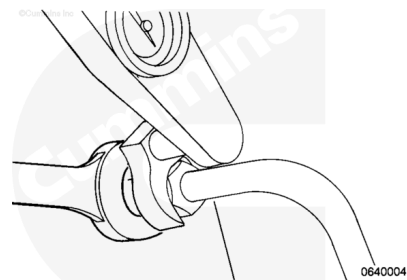
Connect the fuel supply line to the control valve body.



It is a good service practice to use two wrenches when tightening the fuel fittings.

Support the fitting. Use a crowfoot wrench and a torque wrench to tighten the fuel supply line nut.

Torque Value: 120 n•m [89 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

Install the electronic control module (ECM). Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html) in the Troubleshooting and Repair Manual, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.

